

```
package com.mycompany.login;
```

```
import java.util.Scanner;
```

```
public class Login {
```

```
    //THIS HOLDS THE USERS PERSONAL DETAILS //
```

```
    private String firstName;
```

```
    private String lastName;
```

```
    private String username;
```

```
    private String password;
```

```
    private String cellPhoneNumber;
```

```
    //THE CONSTRUCTOR//
```

```
    public Login(String firstName, String lastName, String username, String password, String  
cellPhoneNumber) {
```

```
        this.firstName = firstName;
```

```
        this.lastName = lastName;
```

```
        this.username = username;
```

```
        this.password = password;
```

```
        this.cellPhoneNumber = cellPhoneNumber;
```

```
    }
```

```
    // USERNAME CHECK//
```

```
    public boolean checkUserName() {
```

```
    return username.contains("_") && username.length() <= 5;
}
```

```
// PASSWORD CHECK//
```

```
public boolean checkPasswordComplexity() {
    if (password.length() < 8) return false;

    boolean hasCapital = false;
    boolean hasNumber = false;
    boolean hasSpecial = false;

    for (char c : password.toCharArray()) {
        if (Character.isUpperCase(c)) hasCapital = true;
        else if (Character.isDigit(c)) hasNumber = true;
        else if (!Character.isLetterOrDigit(c)) hasSpecial = true;
    }

    return hasCapital && hasNumber && hasSpecial;
}
```

```
// CHECKS IF CELLPHONE NUMBER IS CORRECT//
```

```
public boolean checkCellPhoneNumber() {
    String regex = "^\\+27\\d{9}$";
    return cellPhoneNumber.matches(regex);

    //          REFERENCE FOR THE INTENTIONAL CODE +27          //
```

```
    /** Regex sourced and adapted from: https://www.baeldung.com/java-regex-validate-phone-numbers */
```

```
}
```

```
// REGISTRATION MESSAGE//
```

```
public String registerUser() {
```

```
    if (!checkUserName()) {
```

```
        return "Username is not correctly formatted; please ensure that your username contains a underscore and is no more than five characters in length.";
```

```
    }
```

```
    if (!checkPasswordComplexity()) {
```

```
        return "Password is not correctly formatted; please ensure that the password contains atleast 8 characters, a capital letter, a number and a special character.";
```

```
    }
```

```
    if (!checkCellPhoneNumber()) {
```

```
        return "Cell phone number incorrectly formatted or does not contain international code (e.g. +27...).";
```

```
    }
```

```
    return "Cellphone number successfully captured";
```

```
}
```

```
// LOGIN CHECK//
```

```
public boolean loginUser(String enteredUsername, String enteredPassword) {
```

```
    return this.username.equals(enteredUsername) &&  
    this.password.equals(enteredPassword);
```

```
}
```

```
// LOGIN MESSAGE//
```

```
public String returnLoginStatus(String enteredUsername, String enteredPassword) {
```

```
    if (loginUser(enteredUsername, enteredPassword)) {
```

```
        return "Welcome " + firstName + " " + lastName + ", great to see you again!";
```

```
    } else {
```

```
        return "Username or password is incorrect, please try again";
```

```
    }
```

```
}
```

```
////          MAIN          ////
```

```
public static void main(String[] args) {
```

```
    // USER WITH ACCOUNT//
```

```
    Login user = new Login("Kyle", "Jackson", "kyl_1", "Ch&&sec@ke99!",  
"+27838968976");
```

```
    showMenu(user);
```

```
}
```

```
// MENU HELPER  KEEPS REGISTERED USER ALIVE WHEN RETURNING TO WELCOME  
SCREEN//
```

```
public static void showMenu(Login user) {
```

```
    Scanner scanner = new Scanner(System.in);
```

```
System.out.println("\n WELCOME TO QUICKCHAT");

System.out.println("1. Register New User");

System.out.println("2. Login");

System.out.println("3. Exit");

System.out.print("Choose option: ");

int choice = scanner.nextInt();

scanner.nextLine();


switch (choice) {

    case 1:

        // REGISTER NEW USER//

        System.out.println("\n REGISTRATION FOR NEW USERS");


        System.out.print("First Name: ");

        String firstName = scanner.nextLine();


        System.out.print("Last Name: ");

        String lastName = scanner.nextLine();


        System.out.print("Username: ");

        String username = scanner.nextLine();


        // CAPTURES USERNAME//

        Login tempUser = new Login(firstName, lastName, username, "", "+270000000000");

        if (tempUser.checkUserName()) {
```

```
        System.out.println("Username successfully captured");
    } else {
        System.out.println("Username is not correctly formatted; please ensure that your
username contains a underscore and is no more than five characters in length.");
    }

    System.out.print("Password: ");

    String password = scanner.nextLine();

    // CAPTURES PASSWORD//

    Login tempUser2 = new Login(firstName, lastName, username, password,
"+270000000000");

    if (tempUser2.checkPasswordComplexity()) {
        System.out.println("Password successfully captured");
    } else {
        System.out.println("Password is not correctly formatted; please ensure that the
password contains atleast 8 characters, a capital letter, a number and a special
character.");
    }

    System.out.print("Cell (+27...): ");

    String cell = scanner.nextLine();

    // CAPTURES CELLPHONE NUMBER//

    Login newUser = new Login(firstName, lastName, username, password, cell);

    if (newUser.checkCellPhoneNumber()) {
        System.out.println("Cellphone number successfully captured");
    }
}
```

```
    } else {  
        System.out.println("Cell phone number incorrectly formatted or does not contain  
international code (e.g. +27...).");  
    }
```

```
    // SAVES NEW USER IF VALID//
```

```
    if (newUser.checkUserName() && newUser.checkPasswordComplexity() &&  
newUser.checkCellPhoneNumber()) {
```

```
        user = newUser;
```

```
        System.out.println("User saved. You can now login.");
```

```
    } else {
```

```
        break;
```

```
    }
```

```
    // GO BACK TO WELCOME SCREEN//
```

```
    showMenu(user);
```

```
    break;
```

```
case 2:
```

```
    // LOGIN//
```

```
    System.out.println("\n USER LOGIN");
```

```
    System.out.print("Username: ");
```

```
    String enteredUsername = scanner.nextLine();
```

```
    System.out.print("Password: ");
```

```
String enteredPassword = scanner.nextLine();
```

```
System.out.println(user.returnLoginStatus(enteredUsername, enteredPassword));
```

```
break;
```

```
case 3:
```

```
    System.out.println("Goodbye ");
```

```
    break;
```

```
default:
```

```
    System.out.println("Invalid option.");
```

```
}
```

```
scanner.close();
```

```
}
```

```
}
```